HP-200 Hydraulic Press — Operation & Maintenance Manual

Fictional equipment created for software demonstration only. Manufacturer: Example Press Works (fictional). Document rev. C.

1. General safety

WARNING: Only trained operators may run the HP-200. Wear safety glasses, cut-resistant gloves and steel-toe shoes in the press area. Never place hands between the ram and the bed while the hydraulic unit is energised. Both light curtains must be tested at the start of every shift by passing the test rod through the beam; the ram must stop within 120 ms.

Before any maintenance, cleaning, or die change, perform the Lockout/Tagout (LOTO) procedure in section 3 and release stored hydraulic pressure.

2. Technical specifications

Item	Value
Maximum force	200 kN
Ram stroke	250 mm
Working pressure	180 bar (max. relief valve setting 195 bar)
Hydraulic oil	ISO VG 46 hydraulic oil, tank capacity 60 L
Normal oil temperature	35–55 °C
Motor	7.5 kW, 400 V, 3-phase, 50 Hz
Oil filter	Return-line filter cartridge, part no. HF-46-10 (10 µm)
Light curtain	Type 4, stopping time limit 120 ms

3. Lockout/Tagout (LOTO) procedure

1) Tell affected operators that the press will be shut down. 2) Return the ram to top-dead-centre and press STOP. 3) Turn the main disconnect switch Q1 to OFF and fit your personal padlock and tag. 4) Open the pressure-release valve V3 on the hydraulic unit and wait until gauge G1 reads 0 bar. 5) Place the safety block under the ram. 6) Try to start the press to verify it does not move (zero-energy check). Only the person who fitted the lock may remove it.

4. Error codes

Code	Meaning	Action
E-01	Emergency stop pressed	Release the E-stop, find the reason it was pressed, press RESET.
E-03	Light curtain interrupted during stroke	Remove the obstruction. If it repeats with no obstruction, clean the lenses and check alignment (section 5.2).
E-07	Hydraulic oil temperature high (above 65 °C)	Stop production and let the unit cool. Check oil level, clean the oil cooler fins, check the cooling fan. If E-07 returns within 1 hour, replace the oil filter HF-46-10 (section 6) and call maintenance.
E-12	Low hydraulic pressure (below 150 bar)	Check oil level, look for leaks at hose fittings, check the pump coupling. Do not raise the relief valve above 195 bar.
E-15	Ram position sensor fault	Perform LOTO, check the sensor cable connector X12. Replace sensor if the fault remains.
E-21	Oil filter clogged (differential pressure switch)	Replace filter cartridge HF-46-10 within the current shift.

5. Troubleshooting

5.1 Ram moves slowly: usually low oil level, a clogged filter (see E-21) or air in the hydraulic system. Bleed air by running 10 empty strokes at low speed.

5.2 Light curtain false trips: clean both lenses with a soft lint-free cloth. The alignment LED must be steady green; adjust the brackets until it is.

5.3 Abnormal noise from the pump: a high-pitched whine often means cavitation caused by low oil or a blocked suction strainer. Stop the press and check oil level first.

5.4 Oil leak at cylinder: perform LOTO, release pressure, then tighten the fitting to 45 N·m. If the rod seal leaks, the seal kit SK-200 must be replaced by qualified maintenance staff.

6. Preventive maintenance schedule

Interval	Task
Every shift	Test light curtains; check oil level sight glass; look for leaks.
Weekly	Clean oil cooler fins with compressed air (max 4 bar); check hoses for wear.
Every 500 operating hours	Replace return-line filter cartridge HF-46-10.
Every 2,000 operating hours or yearly	Change hydraulic oil (ISO VG 46, 60 L); take an oil sample.
Yearly	Verify light-curtain stopping time ≤ 120 ms with a stop-time meter; inspect ram guide gibs.

Replacing the oil filter: perform LOTO (section 3), place a drip tray, unscrew the filter bowl, replace the HF-46-10 cartridge and O-ring, hand-tighten plus 1/4 turn, then run 5 empty strokes and check for leaks.

7. Die change

1) Perform LOTO (section 3) and place the safety block. 2) Loosen the four upper die clamps C1–C4 and the lower clamps C5–C8. 3) Use the die cart to slide the die out; never lift dies above 25 kg by hand. 4) Clean the bed and ram faces. 5) Insert the new die, centre it with the locating pins and torque clamps to 60 N·m in a cross pattern. 6) Remove the safety block and lock, then run one slow single stroke to check die closing height.